

Model improvements – Lesson overview

Model improvements are crucial in data science for enhancing the **accuracy**, **efficiency**, and **generalisability** of predictive models. These advancements enable practitioners to tackle **complex challenges** in classification, dealing with **imbalanced datasets**, and optimising **model parameters** for superior outcomes.

In this lesson, we will cover a range of techniques, from **feature subset selection** and **hyperparameter tuning** to addressing **class imbalance** and introducing multiclass classification strategies. We'll be equipped with the knowledge and skills to refine logistic regression models, utilise advanced tuning methods, and effectively manage dataset peculiarities for improved model performance.



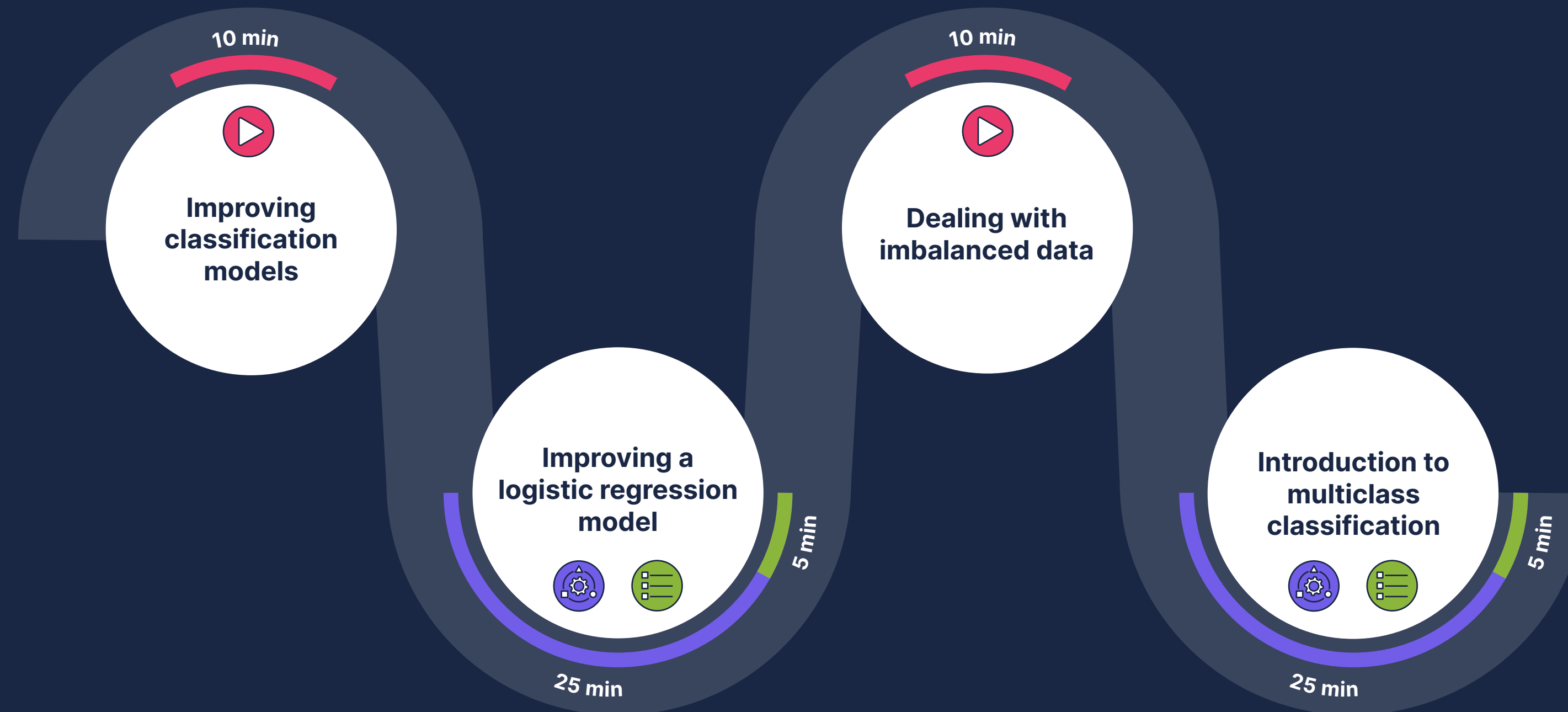
Video



Knowledge questions



Examples



Learning objectives

- Understand and apply feature subset selection techniques such as variance threshold, select KBest, and forward/backward stepwise selection.
- Master hyperparameter tuning for various models, including logistic regression, Random Forest, and Neural Networks.
- Address class imbalance using threshold adjustments and resampling methods.
- Rebuild and evaluate logistic regression models using performance metrics to enhance prediction accuracy.
- Navigate the complexities of multiclass classification.
- Explore various metrics and concepts used for evaluating classification models.

